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Question

The eigen-values of a real symmetric matrix are always

- a) positive
- b) imaginary
- c) real
- d) complex conjugate pairs

Theoretical solution

Let $\mathbf{A}$ be a real symmetric matrix, meaning $\mathbf{A} = \mathbf{A}^\top$ and its entries are real. Let λ be an eigenvalue and $\mathbf{v}$ its eigenvector. The eigenvalue equation is:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (0.1)$$

Taking the complex conjugate of the equation:

$$(\mathbf{A}\mathbf{v})^c = (\lambda\mathbf{v})^c \implies \mathbf{A}^c\mathbf{v}^c = \lambda^c\mathbf{v}^c \quad (0.2)$$

Since $\mathbf{A}$ is a real matrix, $\mathbf{A}^c = \mathbf{A}$. The equation becomes:

$$\mathbf{A}\mathbf{v}^c = \lambda^c\mathbf{v}^c \quad (0.3)$$

Theoretical solution

First, left-multiply the original eigenvalue equation by $(\mathbf{v}^c)^\top$:

$$(\mathbf{v}^c)^\top \mathbf{A} \mathbf{v} = (\mathbf{v}^c)^\top (\lambda \mathbf{v}) = \lambda ((\mathbf{v}^c)^\top \mathbf{v}) \quad (0.4)$$

Next, take the transpose of equation (0.3):

$$(\mathbf{A} \mathbf{v}^c)^\top = (\lambda^c \mathbf{v}^c)^\top \implies (\mathbf{v}^c)^\top \mathbf{A}^\top = \lambda^c (\mathbf{v}^c)^\top \quad (0.5)$$

Since $\mathbf{A}$ is symmetric, $\mathbf{A}^\top = \mathbf{A}$.

$$(\mathbf{v}^c)^\top \mathbf{A} = \lambda^c (\mathbf{v}^c)^\top \quad (0.6)$$

Theoretical solution

Now, right-multiply this equation by $\mathbf{v}$:

$$(\mathbf{v}^c)^\top \mathbf{A} \mathbf{v} = \lambda^c ((\mathbf{v}^c)^\top \mathbf{v}) \quad (0.7)$$

By comparing equations (0.4) and (0.7), we see that:

$$\lambda ((\mathbf{v}^c)^\top \mathbf{v}) = \lambda^c ((\mathbf{v}^c)^\top \mathbf{v}) \quad (0.8)$$

$$(\lambda - \lambda^c) ((\mathbf{v}^c)^\top \mathbf{v}) = 0 \quad (0.9)$$

Theoretical solution

The term $(\mathbf{v}^c)^\top \mathbf{v}$ is the squared norm of $\mathbf{v}$:

$$(\mathbf{v}^c)^\top \mathbf{v} = \sum_i v_i^c v_i = \sum_i |v_i|^2 = \|\mathbf{v}\|^2 \quad (0.10)$$

Since $\mathbf{v}$ is an eigenvector, it cannot be the zero vector, so its norm $\|\mathbf{v}\|^2$ must be a positive real number.

Given that $\|\mathbf{v}\|^2 \neq 0$, the only way for the equation $(\lambda - \lambda^c)\|\mathbf{v}\|^2 = 0$ to hold is if:

$$\lambda - \lambda^c = 0 \implies \lambda = \lambda^c \quad (0.11)$$

A number is equal to its own conjugate only if its imaginary part is zero. Therefore, λ must be real. The correct option is (c).